

Question B — The Bias of the Independence Assumption Between Trees

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The M trees of the forest share the same data stream — in which direction does this bias the article's formula, which assumes the τ_i independent to compute $\tau_{\text{ARF}} = \min_i \tau_i$? New objects relative to Question A : S is the complete realization of the stream (one seed) ; U_i the randomness specific to tree i (Poisson weights drawn online for bagging, random feature subspaces) ; $F(s)$ the marginal, unconditional law of a single tree ; $G_S(s) := P(\tau_1 > s \mid S)$ its survival function *given* the stream ; $M = 10$ the number of trees in the R2 campaign.

1 Conditional Factorization

Once S is fixed (the entire trajectory (x_t, y_t) known), what still distinguishes each tree is U_i : the Poisson weights drawn online for bagging (Oza & Russell, 2001 — this is not classical bagging by resampling a fixed dataset, which is impossible in streaming, but a random weight assigned to each new observation), and the random feature subspaces tested at each node. These draws are, by construction, independent across trees : tree i never sees tree j 's weights, and a tree newly created at replacement inherits no information (neither weights nor structure) from the tree it replaces — the only link between the two is temporal, the new one having started learning more recently.

The adaptation time of tree i is a deterministic function of the pair (S, U_i) . Once S is fixed, only the variation in U_i remains, and the U_i are i.i.d. across trees. Hence, conditionally on S , the τ_i are indeed independent (and identically distributed, since all trees share the same procedure).

Result. Defining the conditional survival function $G_S(s) := P(\tau_1 > s \mid S)$, conditional independence gives

$$P(\tau_{\text{ARF}} > s \mid S) = P(\forall i, \tau_i > s \mid S) = \prod_{i=1}^M P(\tau_i > s \mid S) = G_S(s)^M.$$

2 The Direction of the Bias, via Convexity

We uncondition using the law of total probability :

$$P(\tau_{\text{ARF}} > s) = \mathbb{E}_S[G_S(s)^M].$$

The function $x \mapsto x^M$ is convex on $[0, 1]$ for $M \geq 1$. Jensen's inequality gives directly

$$\mathbb{E}_S[G_S(s)^M] \geq (\mathbb{E}_S[G_S(s)])^M = (1 - F(s))^M,$$

since $\mathbb{E}_S[G_S(s)] = P(\tau_1 > s) = 1 - F(s)$ (the marginal law, unconditioned).

Result.

$$P(\tau_{\text{ARF}} > s) \geq (1 - F(s))^M \iff P(\tau_{\text{ARF}} \leq s) \leq 1 - (1 - F(s))^M.$$

The true probability of having already adapted by time s is always less than or equal to what the independence formula predicts : the manuscript **overestimates the repair speed** — τ_{ARF} is stochastically larger (slower) than the article claims.

Equality condition. Jensen’s inequality is an equality if and only if $G_S(s)$ is almost surely constant — that is, if the difficulty of adaptation does not actually depend on the specific stream realization. Physically, this case essentially never occurs here : every seed produces a different noise trajectory after the drift, so $G_S(s)$ genuinely varies across realizations. Independence is therefore never exact — only an approximation, systematically optimistic.

3 Operational Consequence

A real repair that is slower than expected wins the race against the detector less often than the independence formula predicts. In two steps :

Quantity	Effect of independence
Adaptation speed (τ_{ARF})	Too optimistic — predicts faster than reality
P_{miss}	Too pessimistic — overestimates the blind-spot risk

Both hold simultaneously, with no contradiction : these are two different quantities, linked by the fact that a slower repair gives the detector more chances.

M_{crit} . M_{crit} is defined via the CDF under independence. Since the true CDF is everywhere less than or equal to it (Section 2), the true M needed to reach a given confidence level β is greater than or equal to M_{crit} .

Result. The corollary is not invalidated — it is **reclassified as a conservative certificate** : “at least M_{crit} trees are needed” remains true, but it is no longer an exact estimate of the sufficient number. It is a valid lower bound, not a tight answer.

4 The Correlation Trap

Toy case $M = 2$ on $\{1, 2\}$: is an anti-monotone coupling compatible with the real ARF?

The counterexample. Support $\{1, 2\}$, marginals $P(\tau_i = 1) = \frac{1}{2}$. Anti-monotone coupling : $\tau_1 = 1 \iff \tau_2 = 2$ (perfectly opposite). Under this coupling, $\{\tau_1 = 2\}$ implies $\{\tau_2 = 1\}$, so $P(\tau_1 > 1, \tau_2 > 1) = P(\tau_1 = 2, \tau_2 = 2) = 0$, hence

$$\text{Cov}(\mathbf{1}\{\tau_1 > 1\}, \mathbf{1}\{\tau_2 > 1\}) = 0 - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4}.$$

Under independence this covariance equals 0 : the anti-monotone coupling thus yields a strictly negative value.

Why this is impossible in the real ARF. Reusing the factorization from Section 1 and the law of total covariance

$$\text{Cov}(X, Y) = \mathbb{E}_S[\text{Cov}(X, Y \mid S)] + \text{Cov}(\mathbb{E}[X \mid S], \mathbb{E}[Y \mid S]),$$

with $X = \mathbf{1}\{\tau_1 > 1\}$, $Y = \mathbf{1}\{\tau_2 > 1\}$: the first term is zero (conditional independence given S , Section 1). The second is computed directly, since $\mathbb{E}[X \mid S] = \mathbb{E}[Y \mid S] = G_S(1)$:

$$\text{Cov}(\mathbf{1}\{\tau_1 > 1\}, \mathbf{1}\{\tau_2 > 1\}) = \text{Var}(G_S(1)) \geq 0 \quad \text{always.}$$

Conclusion — the trap defused. The real ARF can only produce a covariance ≥ 0 between two trees — never negative. The anti-monotone coupling, which requires a covariance of $-1/4$, is therefore structurally impossible in the real model : the dependence shared through S can only go one way (independence if $\text{Var}(G_S) = 0$, positive dependence otherwise), never the other. This confirms and locks in the conclusion of Section 2 : no negative dependence scenario could contradict “independence is too optimistic about speed”.

5 Confrontation with the Experimental Side (Q C.1)

Salomé’s instrumented campaign (`redaction_QC1_hierarchie.tex`) establishes, with no distributional assumption whatsoever, that τ_{ARF} systematically and one-sidedly underestimates the time needed for a genuinely significant fraction of the forest to have been renewed — an optimistic, one-sided bias, verified without exception across the campaign’s 2,000 runs. This is exactly the direction of the bias established here : positive dependence between trees (Section 2, Jensen) implies that the forest’s true adaptation speed is slower than what the article’s independence assumption would predict. The two results are obtained via completely independent routes — one theoretical and data-free, the other empirical and assumption-free — and they converge exactly on the sign of the bias. It is this kind of cross-validation, rather than an isolated answer to each question, that gives the overall diagnosis its value.

6 Outlook

What remains is to measure $G_S(s)$ empirically on the campaign’s runs (inter-seed variance of a single tree’s survival at a fixed s) in order to quantify — not just establish the sign of — the gap between the independence formula and reality. Also compare R6/R9 (single tree, $M = 1$) against R2 (forest, $M = 10$), as suggested by Minato.